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Embedded Systems and IoT lab

Part A write up

Formulas used :

For total resistance in series it is : $R_1 + R_2 + R_3$

For total resistance in a parallel circuit we have : $1/R_{total} = 1/R_1 + 1/R_2 + 1/R_3$ and so forth and so on

Voltage across each resistor in parallel is the same

Current across each resistor in series is the same

Keeping that in mind current I is equal to V/R and Voltage $V = IR$

Power = VI

So therefore for circuit A(series $20\Omega + 30\Omega + 50\Omega$, 125 V supply)

Total current here will be total voltage over total Resistance and from that we can get the voltage per resistor using the formula $V = IR$

Resistor	Current(A)	Voltage Drop(V)	Power (W)
20 Ω	1.25	25	31.25
30 Ω	1.25	37.5	46.875
50 Ω	1.25	62.5	78.125

For circuit B (parallel $20\Omega + 100\Omega + 50\Omega$, 125 V supply)

Total voltage will be the same across each resistor so to get individual current per resistor we shall use V/R

Resistor	Current(A)	Voltage Drop(V)	Power (W)
20 Ω	6.25	125	781.25
100 Ω	1.25	125	156.25

50 Ω	2.5	125	312.5
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